MATH222 Linear Algebra II

Midterm Exam October. 20, 2025

Full Name :	Student Number :

1. (100 Points) Mark each statement true or false.
(Your score is 20 if your answer is correct. Your score is 5 if you don't mark. And if your answer is wrong, then your score is -10.)

(1) (20 Points) For $A \in M_{n\times n}(F)$, A is not invertible if and only if 0 is an eigenvalue of A.	(1) True ( ) False ( )
(2) (20 Points) Let $A \in M_{n\times n}(\mathbb{R})$ be an upper triangular matrix whose diagonal entries are all same. Then A is diagonalizable if and only if A is a diagonal matrix.	(2) True ( ) False ( )
(3) (20 Points) Let $A \in M_{n\times n}(\mathbb{R})$ be a matrix of rank r. Then $\det(A-tI_n)$ is divisible by t^{n-r}.	(3) True ( ) False ( )
(4) (20 Points) An inner product $\langle \cdot, \cdot \rangle : V\times V \to F$ is linear in both components.	(4) True ( ) False ( )
(5) (20 Points) The following is an inner product on $M_{n\times n}(\mathbb{R})$. $\langle A, B\rangle = tr(AB)$	(5) True ( ) False ( )

2. (60 Points) Consider the following $n \times n$ matrix A for some $n \geq 3$:

$$A = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

(1) (30 Points) Regard A as an element of $M_{n\times n}(\mathbb{R})$. Then prove that A is not diagonalizable.

(2) (30 Points) Regard A as an element of $M_{n\times n}(\mathbb{C})$. Then prove that A is diagonalizable.

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3. (40 Points) Let V be an inner product space over $\mathbb{R}$, and let $x, y \in V$. Then prove that $\langle x+y, x-y \rangle = 0$ if and only if $\|x\| = \|y\|$.

4. (40 Points) Find A^{100} where

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{bmatrix}.$$

5. (60 Points) Let $\mathbb{R}^n$ be the standard inner product space of column vectors, and let u and v be two non-zero vectors in $\mathbb{R}^n$ which are not orthogonal. Consider the $n \times n$ matrix

$$A = uv^t$$ where v^t denotes the transpose of v.

(1) (30 Points) Let $\mu := v^t u (\neq 0)$. Then prove that 0 and μ are eigenvalue of A.	(2) (30 Points) Prove that A is diagonalizable.